SQL Query Interview Questions

1.Retrieve employee details along with their department name.

```
Select * from Employee
SELECT
   E.EmployeeID As 'Employee ID',
   E.FirstName As 'First Name',
   E.LastName  As 'LastName',
   E.Email As 'Email',
   E.PhoneNumber As 'PhoneNumber',
   E.HireDate As 'HireDate',
   E.JobTitle As 'JobTitle',
   D.DepartmentName as 'Department Name'
FROM Employee AS E
LEFT JOIN Department AS D
   ON E.DepartmentID = D.DepartmentID
ORDER BY E.EmployeeID DESC;
```

2.Find employees hired after January 1, 2021.

```
Select * from Employee where HireDate >='2021-01-01'
```

3.List departments with more than one employee.

```
Select * from Department where DepartmentID in ( Select DepartmentID
 from Employee Group by DepartmentID having count(DepartmentID) >1)
```

4.Find out the total salary paid (BaseSalary + Bonus - Deductions) for each employee.

```
SELECT
   E.EmployeeID AS 'Employee ID',
   E.FirstName AS 'First Name',
   E.LastName AS 'LastName',
   E.Email AS 'Email',
   E.PhoneNumber AS 'PhoneNumber',
   E.HireDate AS 'HireDate',
   SUM(S.BaseSalary + S.Bonus - S.Deductions) AS 'Net Salary'
FROM Employee E
INNER JOIN Salary S ON E.EmployeeID = S.EmployeeID
GROUP BY
   E.EmployeeID,
   E.FirstName,
   E.LastName,
   E.Email,
   E.PhoneNumber,
   E.HireDate
ORDER BY E.EmployeeID DESC;
```

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5. Find the highest paid employee based on net salary.

```
SELECT TOP 1
   E.EmployeeID,
   E.FirstName,
   E.LastName,
   S.BaseSalary,
   S.Bonus,
   S.Deductions,
   (S.BaseSalary + S.Bonus - S.Deductions) AS NetSalary
FROM Employee E
JOIN Salary S ON E.EmployeeID = S.EmployeeID
ORDER BY NetSalary DESC;
```

6.Write a query to find employees who are also managers of their department

```
   Select e.EmployeeID AS 'Employee ID',
            e.FirstName AS 'First Name',
            e.LastName AS 'LastName',
            e.Email AS 'Email',
            e.PhoneNumber AS 'PhoneNumber',
            e.HireDate AS 'HireDate',
            d.DepartmentName as 'Depart name'
         from Employee e Left Join Department d on e.DepartmentID = d.DepartmentID      where e.JobTitle
like '%Manager%'
```

7. Write a query to find employees who are also managers of their department

```
   SELECT
   D.DepartmentName,
   AVG(E.Salary) AS AverageSalary
FROM Department D
JOIN Employee E ON D.DepartmentID = E.DepartmentID
GROUP BY D.DepartmentName;
```

8.Find employees whose salary is below the department average.

```
SELECT
   E.EmployeeID,
   E.FirstName,
   E.LastName,
   E.Salary,
   D.DepartmentName
FROM Employee E
JOIN Department D ON E.DepartmentID = D.DepartmentID
WHERE E.Salary < (
   SELECT AVG(Salary)
   FROM Employee
   WHERE DepartmentID = E.DepartmentID
);
```

9.Update the bonus for all employees in the Sales department by 10%.

```
UPDATE Salary
```

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```
SET Bonus = Bonus * 1.10
WHERE EmployeeID IN (
   SELECT EmployeeID
   FROM Employee
   WHERE DepartmentID in (Select DepartmentID from Department where DepartmentName='Sales')
);
```

10.Delete salary records older than 1 year from today.

```
DELETE FROM Salary
WHERE PayDate < DATEADD(YEAR, -1, GETDATE());
```

11.Show all employees and their salary details, even if salary is missing

```
Select * from Employee E Left Join Salary S on E.EmployeeID =S.EmployeeID
```

12.Count how many employees work in each department.

```
Select DepartmentName,count(*) as EmployeeID  from Employee E
    left join Department D on E.DepartmentID = D.DepartmentID
    group by E.DepartmentID,D.DepartmentName;
```

13.Find departments with no employees.

Type 1:-

```
Select * from Employee where DepartmentID is Null
```

Type 2 :-

```
SELECT
   D.DepartmentID,
   D.DepartmentName
FROM Department D
LEFT JOIN Employee E ON D.DepartmentID = E.DepartmentID
WHERE E.EmployeeID IS NULL;
```

14.Find employees whose salary is above ₹50,000.

```
Select * from Employee where Salary > 50000
```

15.Get the top 5 highest paid employees.

```
SELECT TOP 5
   E.EmployeeID,
   E.FirstName,
   E.LastName,
   S.BaseSalary,
   S.Bonus,
   S.Deductions,
   (S.BaseSalary + S.Bonus - S.Deductions) AS NetSalary
FROM Employee E
JOIN Salary S ON E.EmployeeID = S.EmployeeID
ORDER BY NetSalary DESC;
```

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Tables

Employe Table

```
CREATE TABLE [dbo].[Employee](
	[EmployeeID] [numeric](18, 0) NOT NULL,
	[FirstName] [varchar](50) NULL,
	[LastName] [varchar](50) NULL,
	[Email] [varchar](100) NULL,
	[PhoneNumber] [varchar](15) NULL,
	[HireDate] [date] NULL,
	[JobTitle] [varchar](50) NULL,
	[DepartmentID] [numeric](18, 0) NULL,
	[Salary] [numeric](18, 0) NULL,
PRIMARY KEY CLUSTERED
(
	[EmployeeID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO

ALTER TABLE [dbo].[Employee]  WITH CHECK ADD FOREIGN KEY([DepartmentID])
REFERENCES [dbo].[Department] ([DepartmentID])
GO
```

Deaprtment Table

```
CREATE TABLE [dbo].[Department](
	[DepartmentID] [numeric](18, 0) NOT NULL,
	[DepartmentName] [varchar](100) NOT NULL,
	[Location] [varchar](100) NULL,
	[ManagerID] [numeric](18, 0) NULL,
PRIMARY KEY CLUSTERED
(
	[DepartmentID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY
= OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
```

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Salary Table

```
CREATE TABLE [dbo].[Salary](
	[SalaryID] [numeric](18, 0) NOT NULL,
	[EmployeeID] [numeric](18, 0) NULL,
	[BaseSalary] [decimal](10, 2) NULL,
	[Bonus] [numeric](18, 0) NULL,
	[Deductions] [numeric](18, 0) NULL,
	[PayDate] [date] NULL,
PRIMARY KEY CLUSTERED
(
	[SalaryID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY
= OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO

ALTER TABLE [dbo].[Salary]  WITH CHECK ADD FOREIGN KEY([EmployeeID])
REFERENCES [dbo].[Employee] ([EmployeeID])
GO
```

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